



Metal plates separated the wheels.

1837: The Analytical Engine

While working on the Difference Engine, Babbage had a better idea for a machine that could calculate anything – not just numbers for mathematical tables. The Analytical Engine was composed of a store (equivalent to memory in a modern computer) and a mill (like a CPU in a modern computer). Inspired by the textile industry, Babbage proposed using punch cards to feed instructions into the engine's steam-powered mill. The engine was designed to add, subtract, multiply, and compare, but it was never built.

1890: Tabulating Machine

American inventor Herman Hollerith invented the Tabulating Machine as an efficient way to compile population census data. An operator punched data into a card, slipped the card into the machine, and then pulled the handle. Wherever there was a hole in the card, an associated dial on the machine increased. All the data for one person was entered at the same time.

A dial displayed the counts for a particular category.



The IBM card

In 1928, American company IBM redesigned the punch card to have 80 columns of 12 potential holes. A light shone on each card as it was fed into the computer. If there was no hole, the light was blocked and the machine read in a 0. If there was a hole, the light shone through and was detected by an optical sensor and the machine read in a 1. Each column of potential holes therefore became a 10-digit binary number.

Part of an IBM punch card

Cards were read column-by-column.

► The numbers in a punch card

The numbers in a punch card have fixed meanings. For example, 0100 could be an instruction to add or compare two numbers.

1	•	1	1	1	•	1	•	1	1	1	•	1	1
2	2	2	•	2	2	•	2	2	•	2	2	•	2
3	•	3	3	•	3	3	•	3	3	•	3	3	3
•	4	4	4	•	4	4	4	•	4	4	•	4	4
5	5	•	5	5	5	5	5	5	•	5	5	5	•

BIOGRAPHY

Babbage and Lovelace

English mathematician Charles Babbage (1791–1871) designed two automated calculating machines. Although his ideas couldn't be built using the technology of the time, his Analytical Engine was the first example of a computer that could be programmed to do a variety of different tasks. English mathematician Ada Lovelace (1815–1852) was the first person to see the enormous potential for the Analytical Engine in fields other than pure calculation, and became the first computer programmer as a result. A working model of Babbage's Difference Engine was built by the London Science Museum in 1991.